



Generation AI

Nurturing Future Innovators

Հեղինակային իրավունքը պատկանում է «Հայաստանի գիտության և տեխնոլոգիաների հիմնադրամ»-ին և ենթակա չէ ազատ տարածման:



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Chapter 4: Training and Learning in ML



Session 1: Overview of ML Models

The objectives of the session:

- Explain the difference between classification and regression problems in machine learning using real-world examples.
- Analyze how simple decision rules, thresholds, and boundaries can be used to separate data into classes.
- Interpret scatter plots and time-series data to identify trends, patterns, and group separations.
- Apply basic reasoning techniques (such as drawing lines or curves) to predict future values from historical data.
- Evaluate different decision boundaries and forecasting models by comparing prediction errors and misclassifications.
- Reflect on how human decision-making strategies relate to machine learning algorithms and automated models.

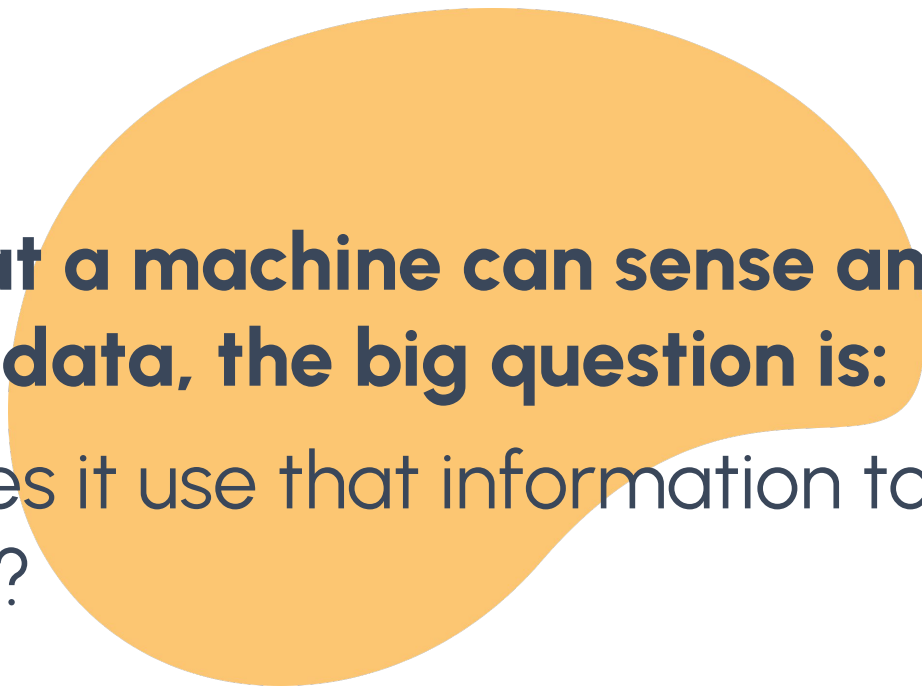
Examine the terms and definitions provided in this chapter following the [link](#).

Key topics discussed in previous chapters

- How machines perceive the world through sensors
- The types of learning (e.g., supervised, unsupervised)
- Types of problems that we can solve with ML (classification, regression, clustering)
- Common challenges in machine learning
- How to process and prepare data for learning.

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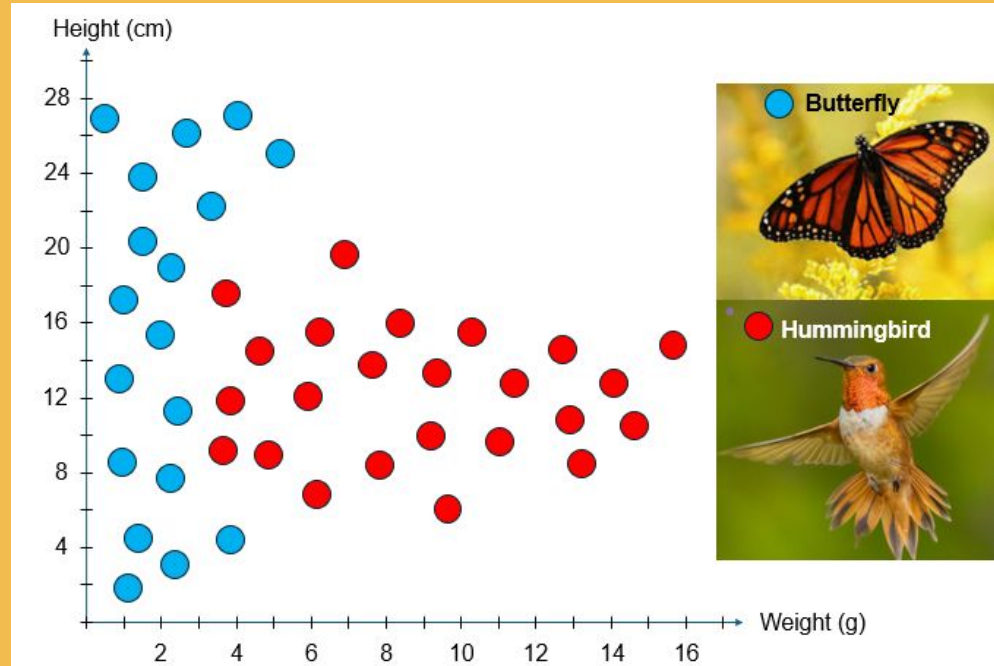


**Now that a machine can sense and
acquire data, the big question is:**

How does it use that information to make a
decision?

Handout

Suppose we conducted a study by observing common flying animals in our garden, where most sightings were of butterflies and hummingbirds. We carefully caught these animals, recorded their height and weight measurements, and then safely released them back into nature. Using this data, we created a height-versus-weight scatter plot to visualize and compare the two groups as shown below.



Key topics discussed in previous chapters

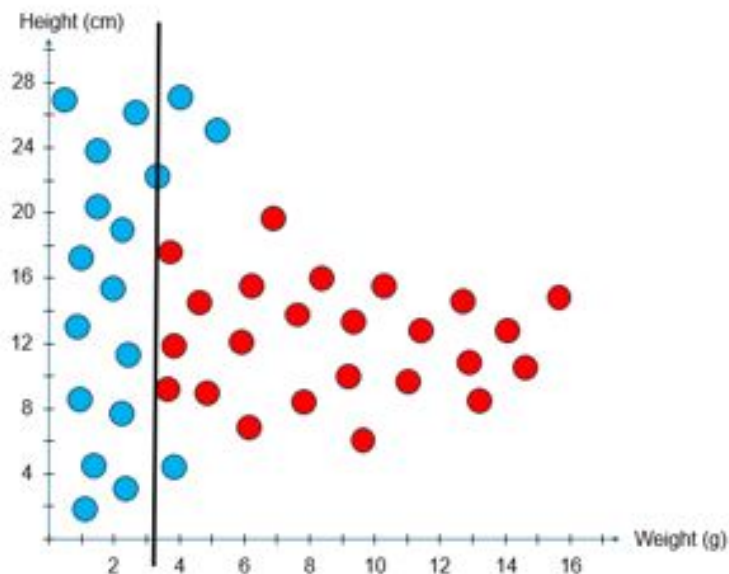
- How machines perceive the world through sensors
- The types of learning (e.g., supervised, unsupervised)
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Identify the rules

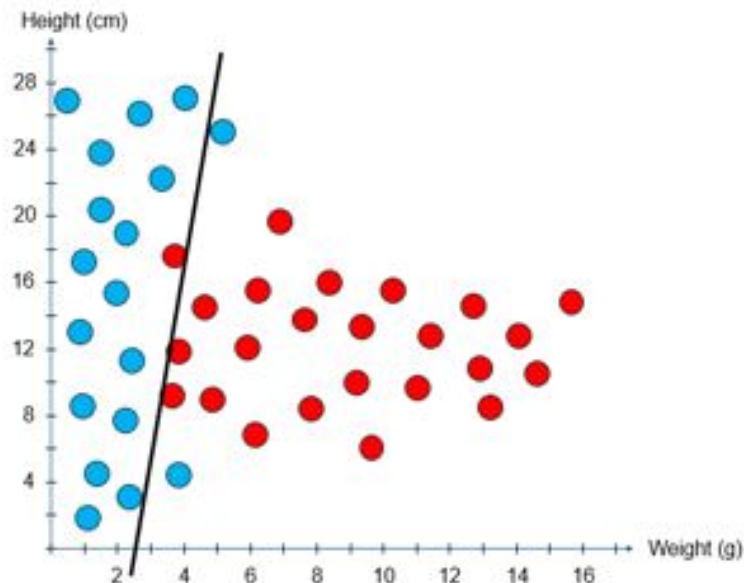
- **Identify a simple rule:** Propose a threshold or condition (using an IF-ELSE statement) that can separate butterflies from hummingbirds based on their height or weight.
- **Linear separation:** Draw a straight line that best separates the two groups on the plot.
- **Nonlinear separation:** Sketch a curved line that more accurately divides the two groups, especially if the data isn't linearly separable.
- **Group boundaries:** Draw a contour or boundary around each group to outline the region where most of their data points lie.

Sample answers

Threshold: if weight < 3 then “butterfly”

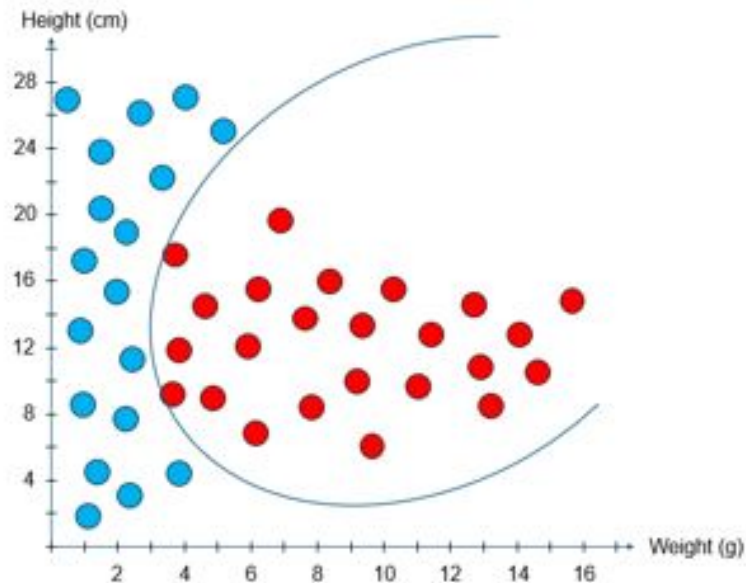


A separating straight line

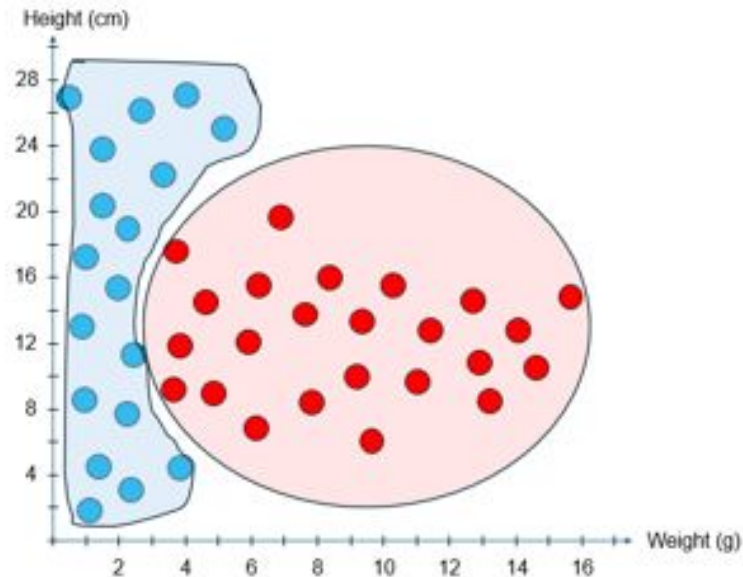


Sample answers

A separating curved line



Group Boundaries



General questions

- ***When choosing a decision boundary, how did you decide on the value?***

Did you try to place it halfway between the two groups to maximize separation? Did you consider how close points were to the boundary or aim to reduce misclassified examples?

- ***When drawing the straight or curved line, what criteria did you use?***

Did you try to minimize the number of points on the wrong side of the line? Did you aim to draw the line so it stayed as far as possible from both groups (i.e., maximum margin)?

General questions

- ***When outlining a boundary around each group, what was your intuition?***

Did you try to include all the points that were close together and exclude outliers? Were you thinking in terms of grouping points based on how near they are to each other (distance-based clusters)?

General questions

Computers use similar ideas when learning to classify data. Different algorithms take different approaches:

- Some adjust thresholds or conditions to split the data,
- Some rotate or shift lines to reduce misclassifications,
- Others use distance-based methods to keep similar points together or cluster them,
- And some draw flexible boundaries that better capture the shape of the data.
- And many other approaches also exist, such as statistical analysis to estimate probabilities, or rule-based systems that rely on expert-defined conditions.
- Other methods use trial and error-based learning.

Regression

Instead of simply separating groups, some tasks require using past observations to predict a future value. This type of task falls under regression or forecasting, which was previously introduced.

- This kind of prediction is common in real-world applications where outcomes are numerical and continuous, rather than categorical.

Examples:

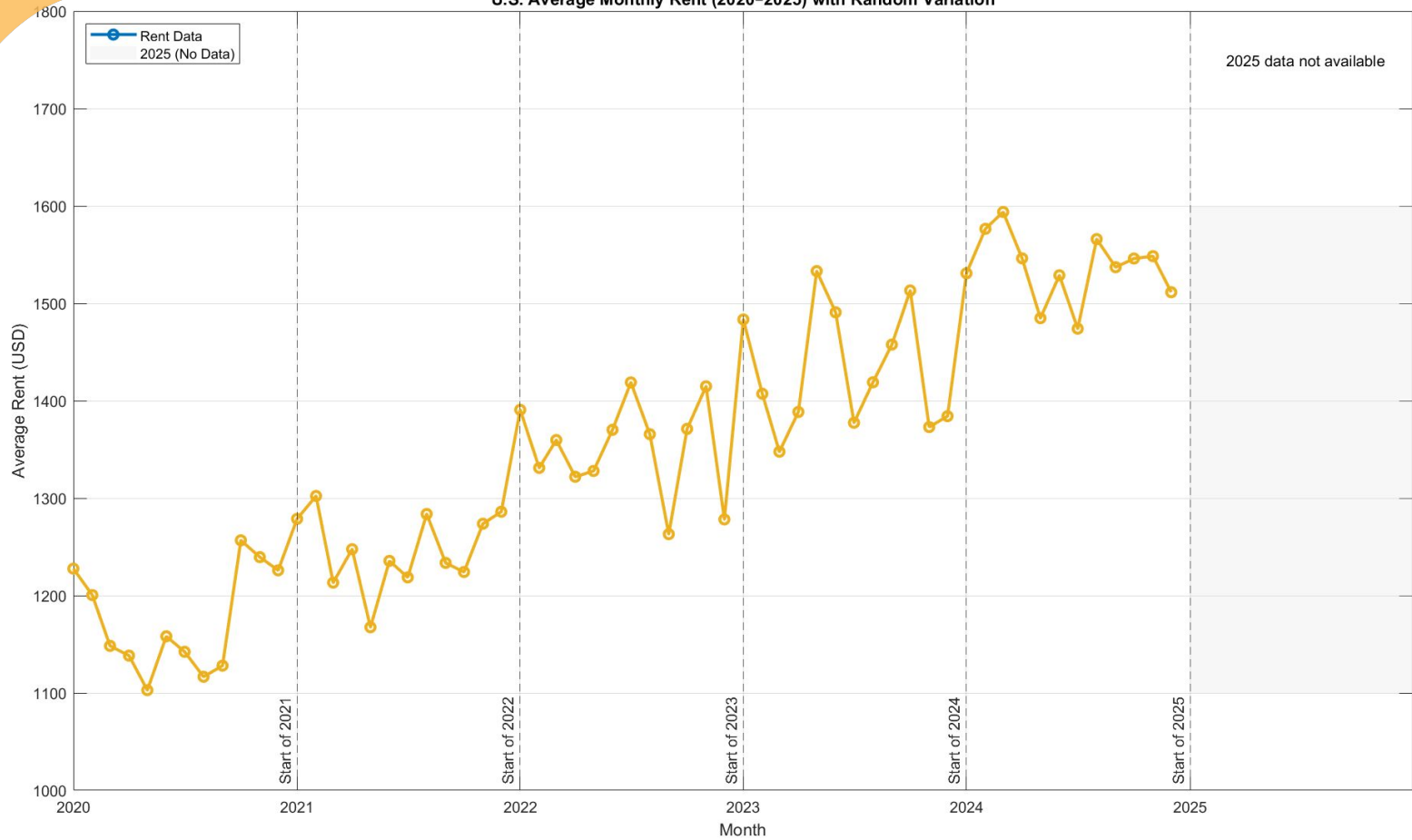
- Predicting the temperature for the next day using past weather data
- Estimating future sales based on historical performance
- Forecasting electricity usage over time in a building or home

Genero

Monthly Average Rent in the U.S. (2020–2025) (Values in \$1000s; 2025 is unknown)

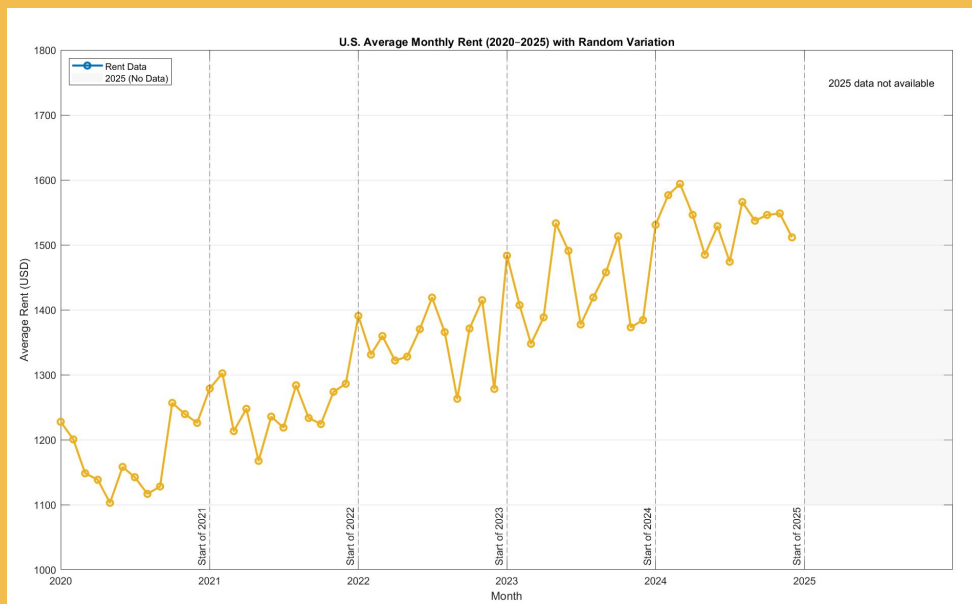
[illegible]

U.S. Average Monthly Rent (2020–2025) with Random Variation

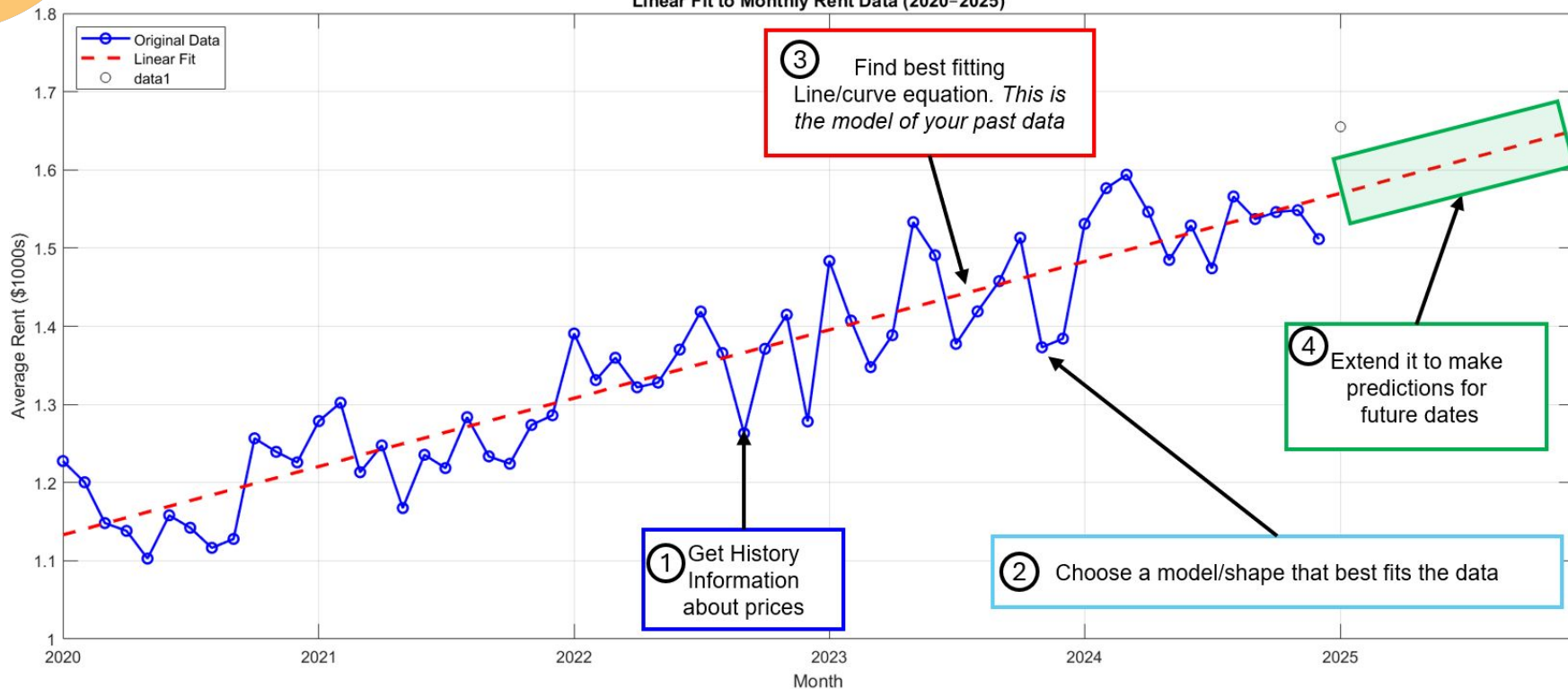


Group Work

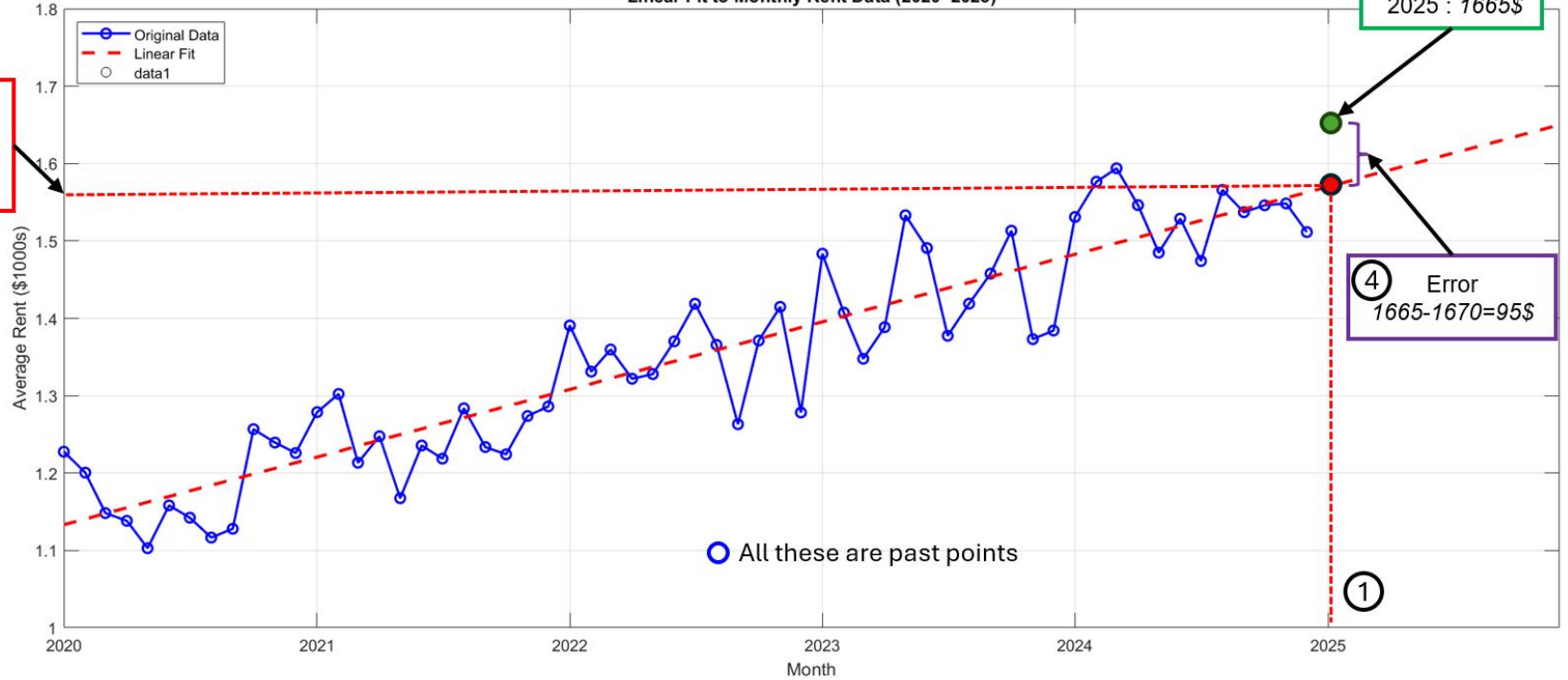
Find the instructions in the file following the [link](#).



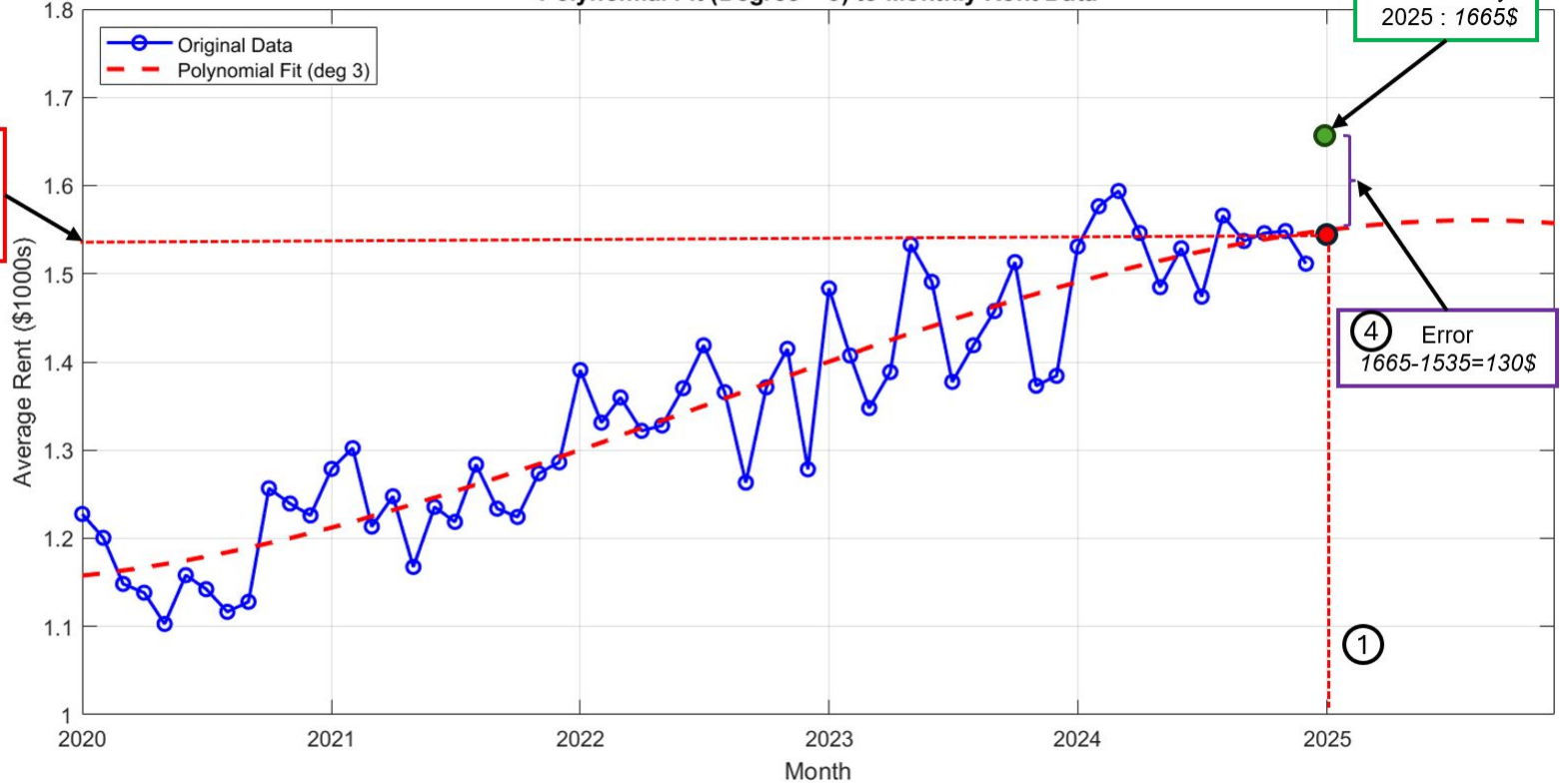
Linear Fit to Monthly Rent Data (2020–2025)



Linear Fit to Monthly Rent Data (2020-2025)



Polynomial Fit (Degree = 3) to Monthly Rent Data



② Predicted Price for January 2025: 1535\$

③ Actual Price For January 2025 : 1665\$

④ Error 1665-1535=130\$

①

What Best Fit actually means

- Did you try to minimize the errors between the actual data and the line?
- Did you simply draw a line that looked like it was midway through the data?
- Did you connect the dots or use a ruler to make the line straight?

There are different ways to define "best", and in data science or machine learning, the best model is often the one that minimizes the overall ***prediction error*** which is a concept that leads naturally into error metrics and model evaluation which is a concept we will study in the next session.

Session 2: Optimization using measures and objective functions

The objectives of the session:

- Explain different distance measures (Euclidean and Manhattan) and their applications in real-world and computational contexts.
- Apply distance formulas to calculate the shortest paths between points using coordinates.
- Understand the term best fit
- Learn how we measure best fit using distances
- Learn distance measures used to calculate the mean residual
- Be able to draw a separating line in a simple classification task.

Examine the terms and definitions provided in this chapter following the [link](#).

Distance measures

- ***Euclidean distance*** – straight-line or geometric distance
- ***Manhattan distance*** – sum of absolute differences
- ***Simple difference*** – used in error calculations like prediction vs. actual

Distance measures are mathematical ways to quantify how far apart two points or values are, and this is crucial for tasks such as classification, clustering, and evaluating prediction accuracy.

**SURREY
CANADA**

Departure

88 Avenue

88 Avenue

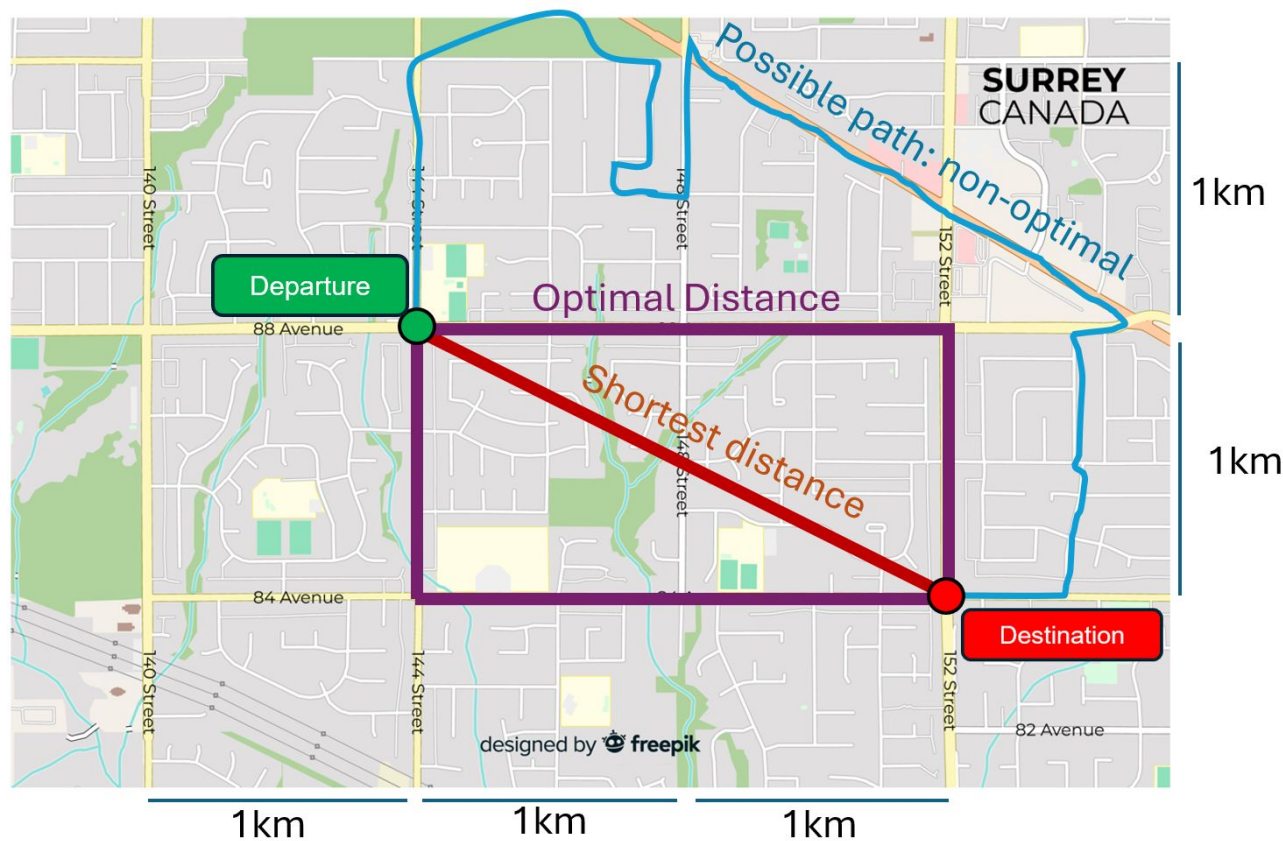
84 Avenue

84 Avenue

Destination

82 Avenue

designed by  **freepik**



- The shortest distance if traveling by car
- The shortest distance of traveling by plane
- Other possible roots from the departure to destination (just estimate or say much larger)

**SURREY
CANADA**

Departure

88 Avenue

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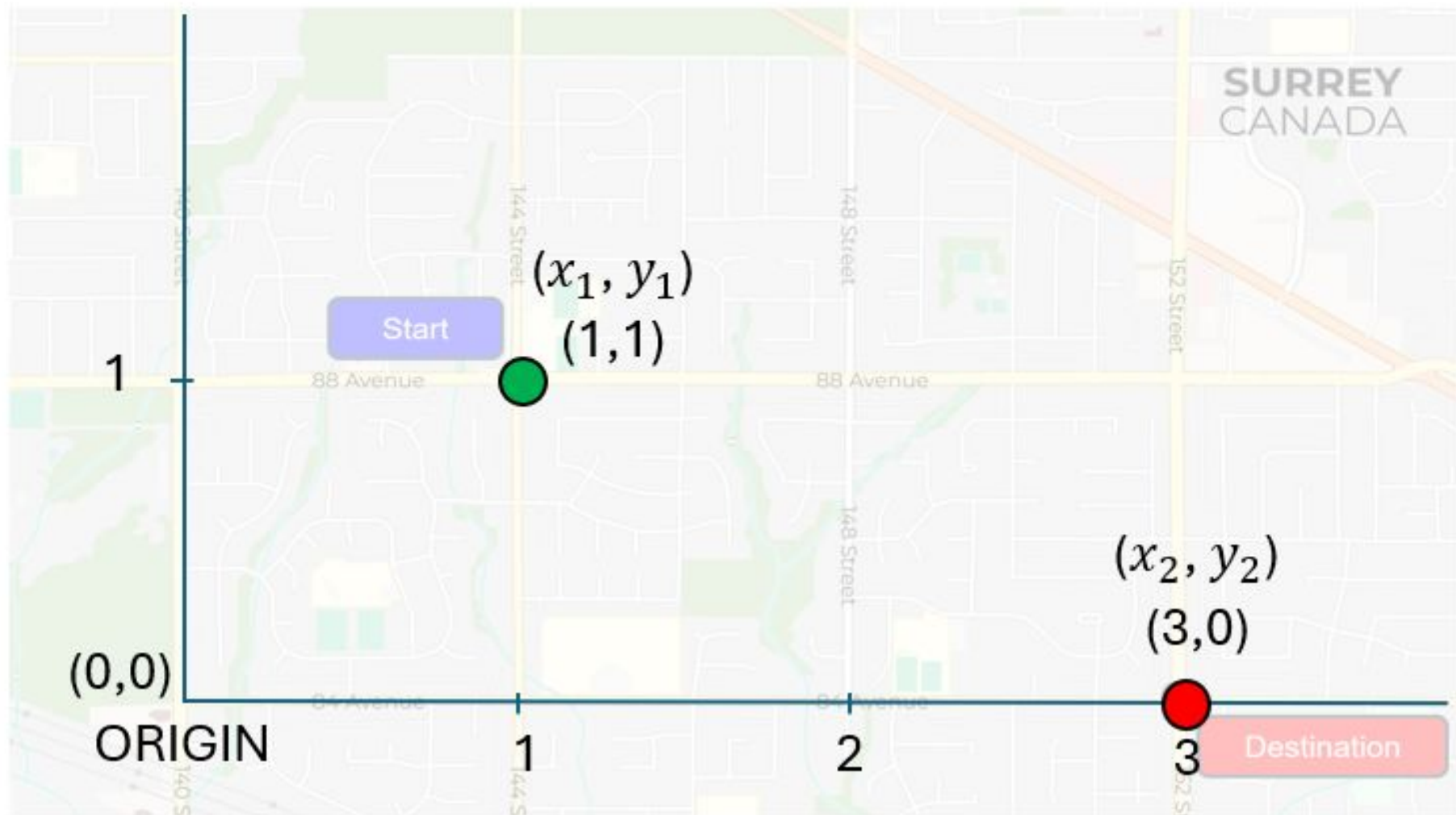
84 Avenue

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Destination

82 Avenue

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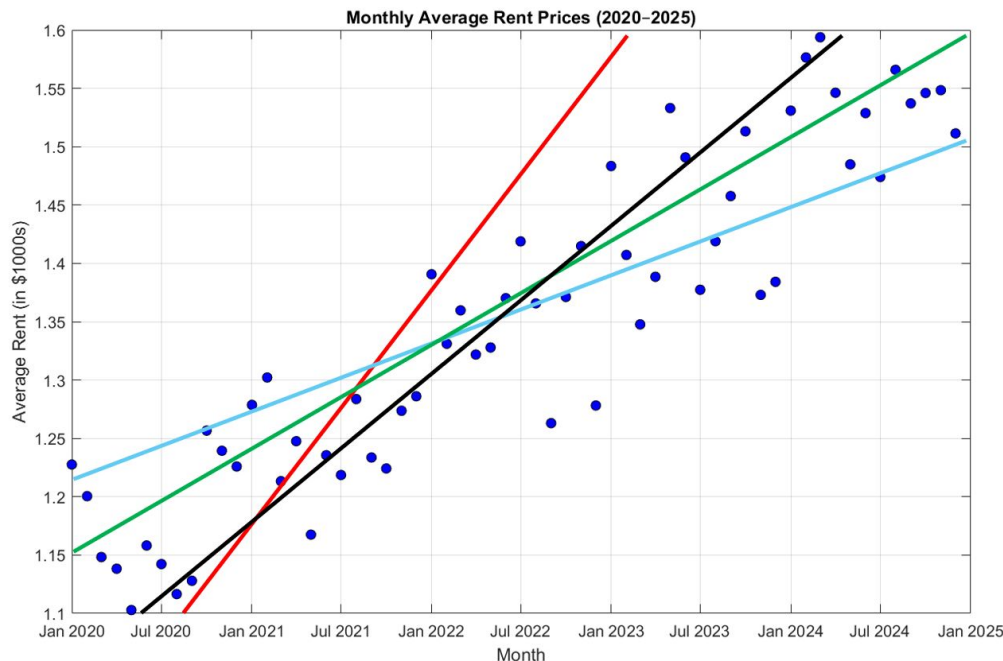
Distance measures

- In the previous example, the main goal was to travel from the departure point to the destination—this was the objective.
- However, the way we reach that objective can vary depending on several factors and constraints. As we observed, if someone were traveling by car or on foot, the best possible path followed the roads and covered a distance of 3 km.
- On the other hand, if traveling by plane or drone, the shortest path was a straight-line distance of approximately 2.23 km. We also looked at a situation where the driver got completely lost, taking an unusual and inefficient route, yet still managed to reach the destination eventually.

Distance measures

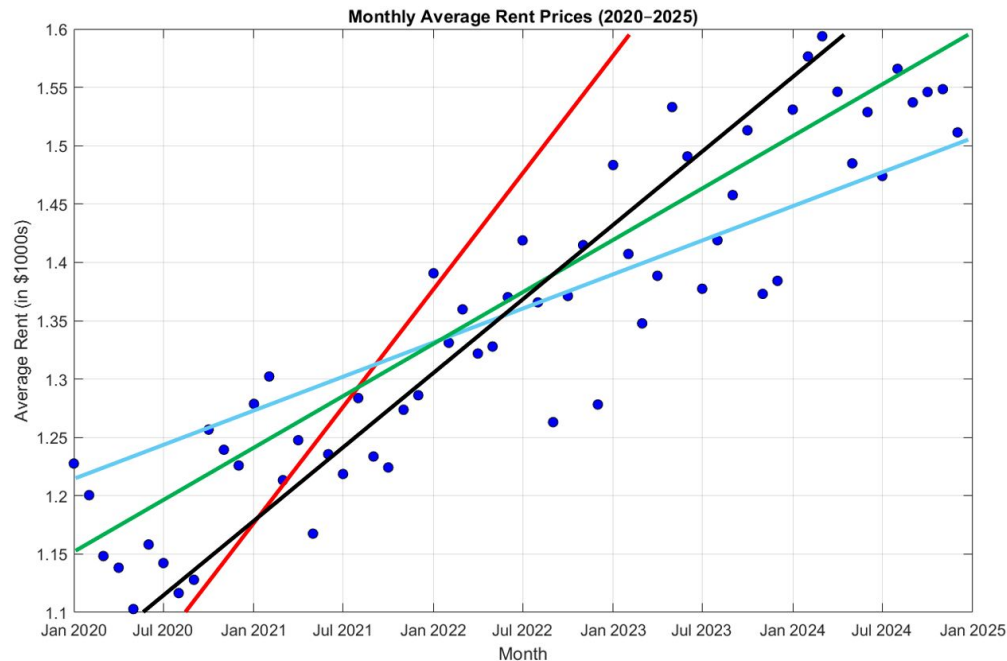
- Just like finding the "best" route, a machine learning model aims to minimize a cost or error.
- ***In regression***, this means reducing the difference between predicted values and actual values—much like minimizing the distance of data points from a fitted line.
- ***In classification***, the objective is to find a boundary that best separates different groups (classes), minimizing misclassifications.

Which one is the best fit?



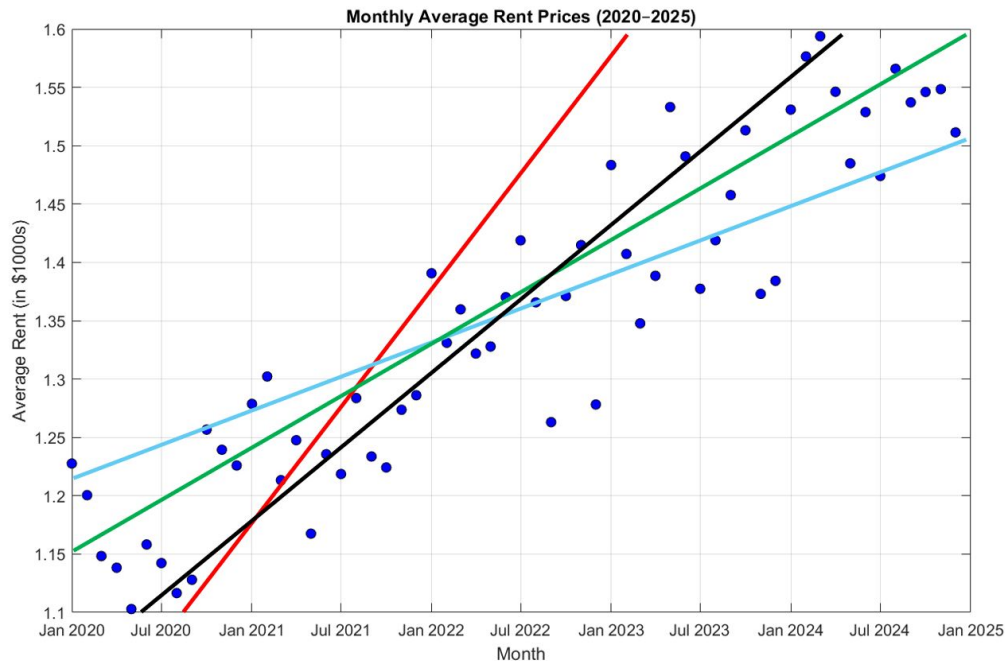
- Does the line follow the general direction of the data?
- Does it stay close to most of the points?
- Are the errors (distances from the points to the line) small and balanced?

The squared errors for blue line



$$\epsilon_{blue} = \frac{\sum d_i^2}{4} = \frac{1.2^2 + 0.8^2 + 1.5^2 + 0.4^2}{4} = \frac{4.49}{4} = 1.225$$

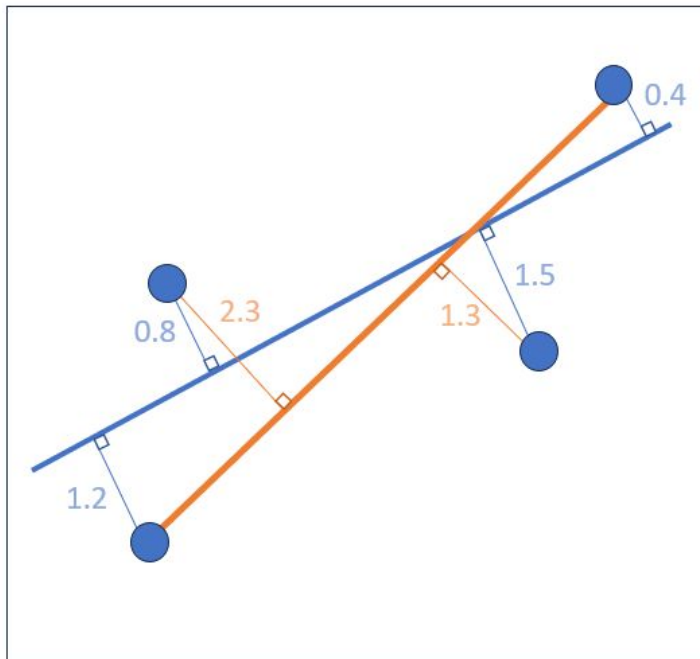
The squared errors for red line



$$\epsilon_{red} = \frac{\sum d_i^2}{4} = \frac{0^2 + 2.3^2 + 1.3^2 + 0^2}{4} = \frac{6.98}{4} = 1.745$$

Example

- Which line has the smallest overall error?

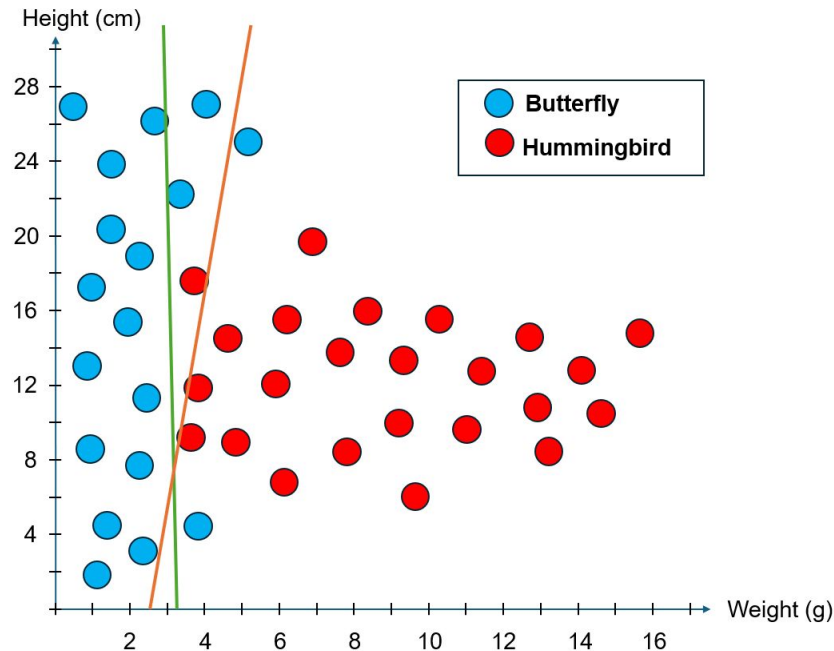


The Key Idea

- Minimizing the objective function is at the heart of machine learning.
- In the context of regression, this means the algorithm searches for the best-fitting line by adjusting its position—shifting and rotating it—to reduce the total error between the line and the actual data points.

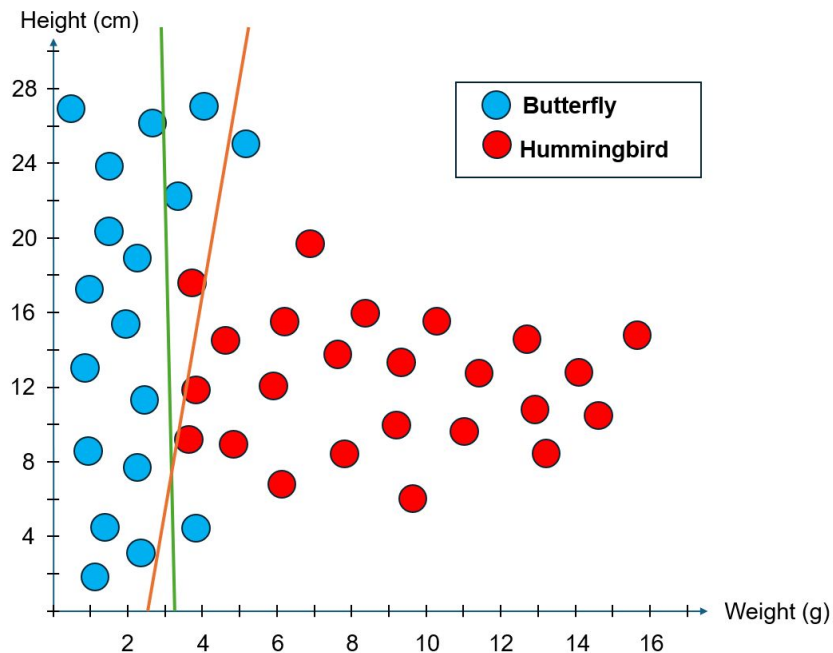
Classification problem

- Let's separate butterflies and hummingbirds based on their height and weight using a scatter plot.
- Each data point represents an animal, and each point is labeled as either blue (18 points for butterfly) or red (24 points for hummingbird).

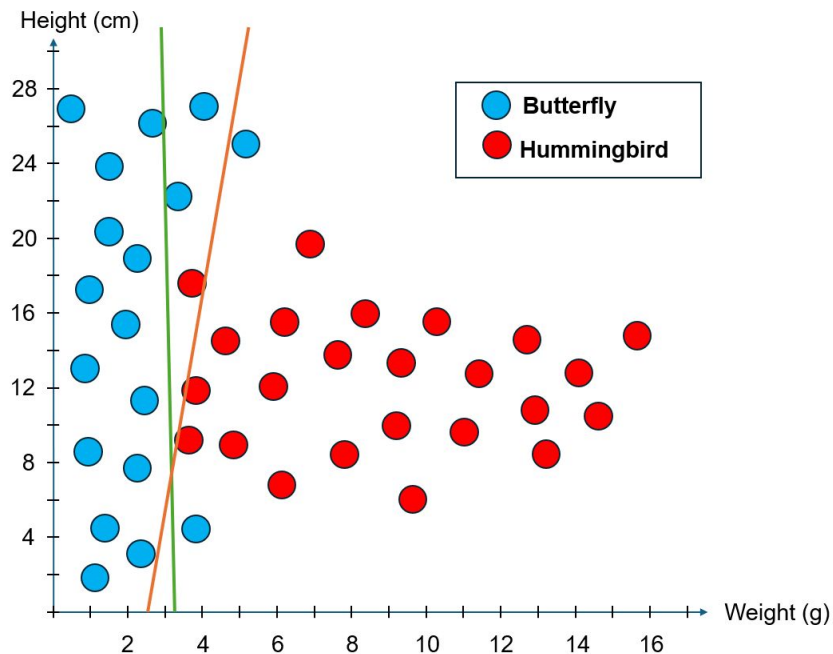


Classification error

- A point is classified correctly if it's on the right side of the line.
- A point is classified incorrectly if it ends up on the wrong side of the line.



Classif



- Green Line: 4 mistakes out of total 42
- Orange Line: 3 mistakes out of total 42

Objective function

- The goal of classification algorithms is to find the line (or curve) that minimizes the classification error (that is, the number of wrongly classified points).
- This is not the only way and that there are more advanced methods for different algorithms.
- But at its core, classification learning tries to draw the best line that separates categories while minimizing the classification error (similar to the overall distance error in regression).

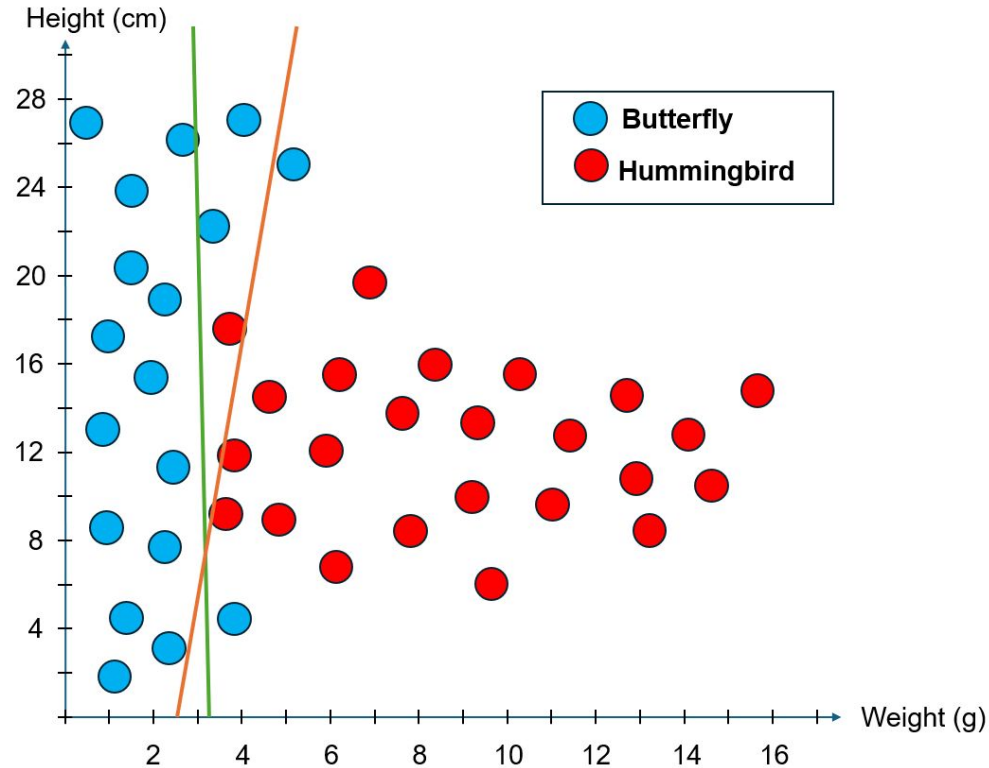
Session 3: Evaluating Model Performance

The objectives of the session:

- Evaluate the model performance using a confusion matrix.
- Identify the strengths and weaknesses of the model based on confusion matrices.
- Define the concept of accuracy of an evaluation.
- Calculate the accuracy of an evaluation.
- Determine the properties of accuracy.
- Explain that having high accuracy does not always mean that the model is performing well.
- Determine the pros and cons of calculating the accuracy.
- Define the concept of recall of an evaluation.
- Calculate the recall of an evaluation.
- Determine the properties of recall.

Examine the terms and definitions provided in this chapter following the [link](#).

Confusion Matrix



Possible Prediction Scenarios

- True Positive (TP): When the model correctly predicts a hummingbird that actually is a hummingbird.
- False Negative (FN): When the model incorrectly predicts a butterfly, but it's actually a hummingbird.
- True Negative (TN): When the model correctly predicts a butterfly that actually is a butterfly.
- False Positive (FP): When the model incorrectly predicts a hummingbird, but it's actually a butterfly.

Actual Class

Positive

Negative

Positive

TP

FP

Negative

FN

TN

Predicted Class

Actual Class

Butterfly

Hummingbird

Butterfly

TP

FP

Hummingbird

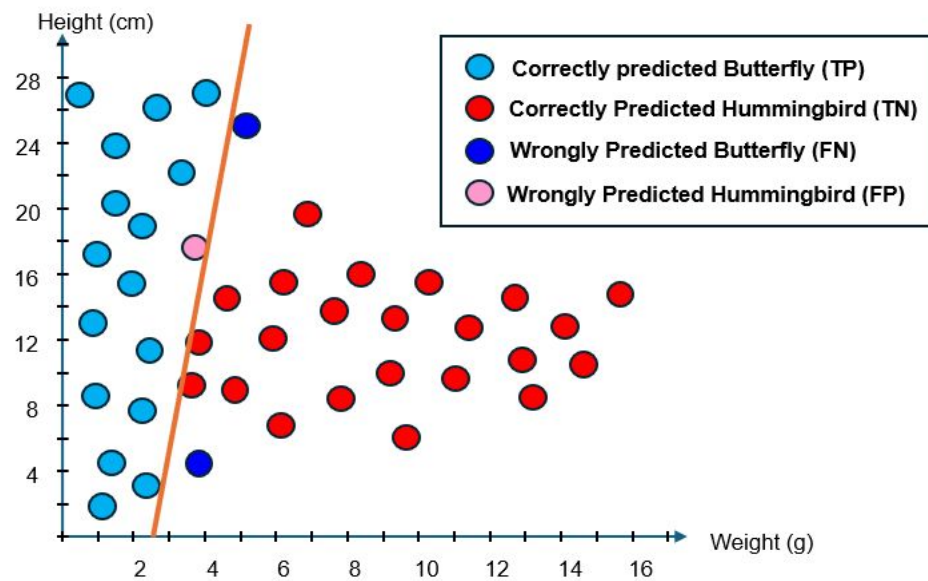
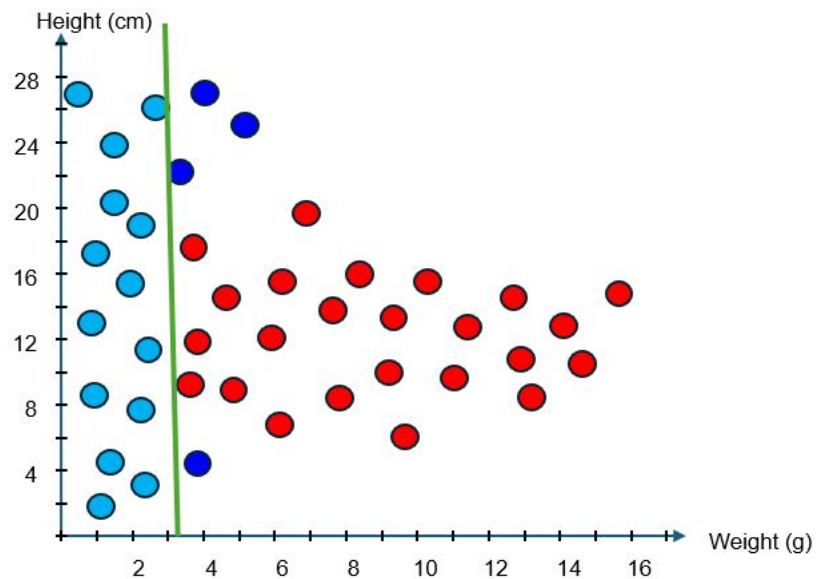
FN

TN

Predicted Class

Tip for remembering:

- "True" or "False" = whether the prediction is correct
- "Positive" or "Negative" = what the model predicted (hummingbird = positive, butterfly = negative)



		Actual Class	
		Positive	Negative
Predicted Class	Positive	14	0
	Negative	4	24

		Actual Class	
		Butterfly	Hummingbird
Predicted Class	Butterfly	16	1
	Hummingbird	2	23

Accuracy

Accuracy tells us how many of the model's predictions were correct out of all the predictions it made.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Number of total predictions}}$$

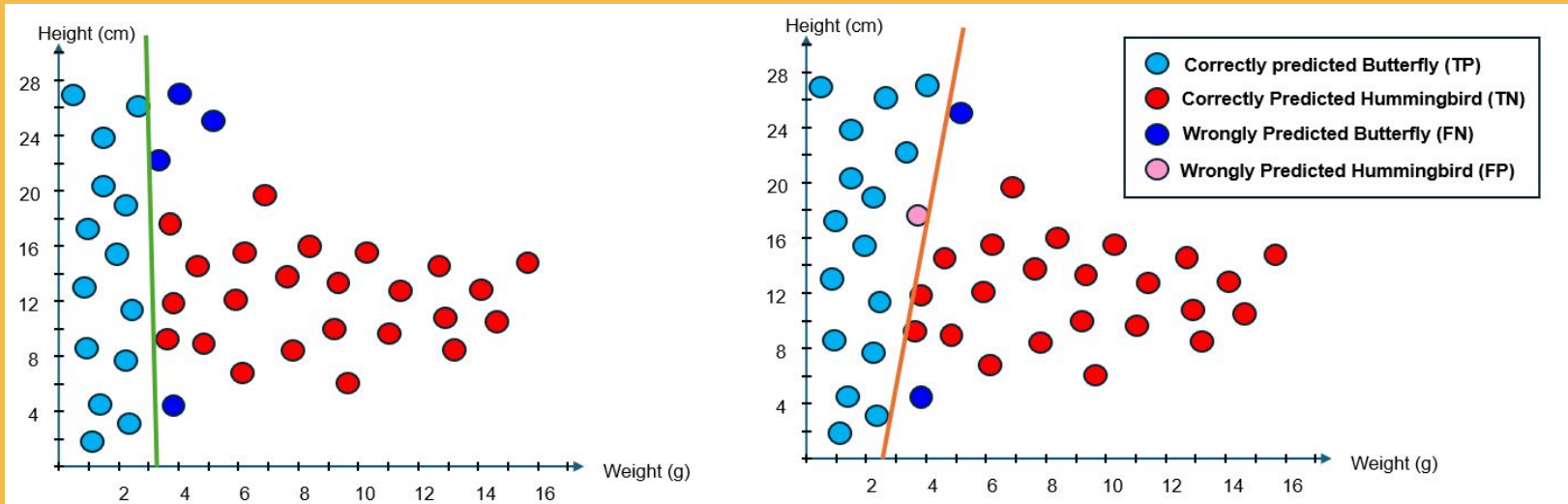
Accuracy

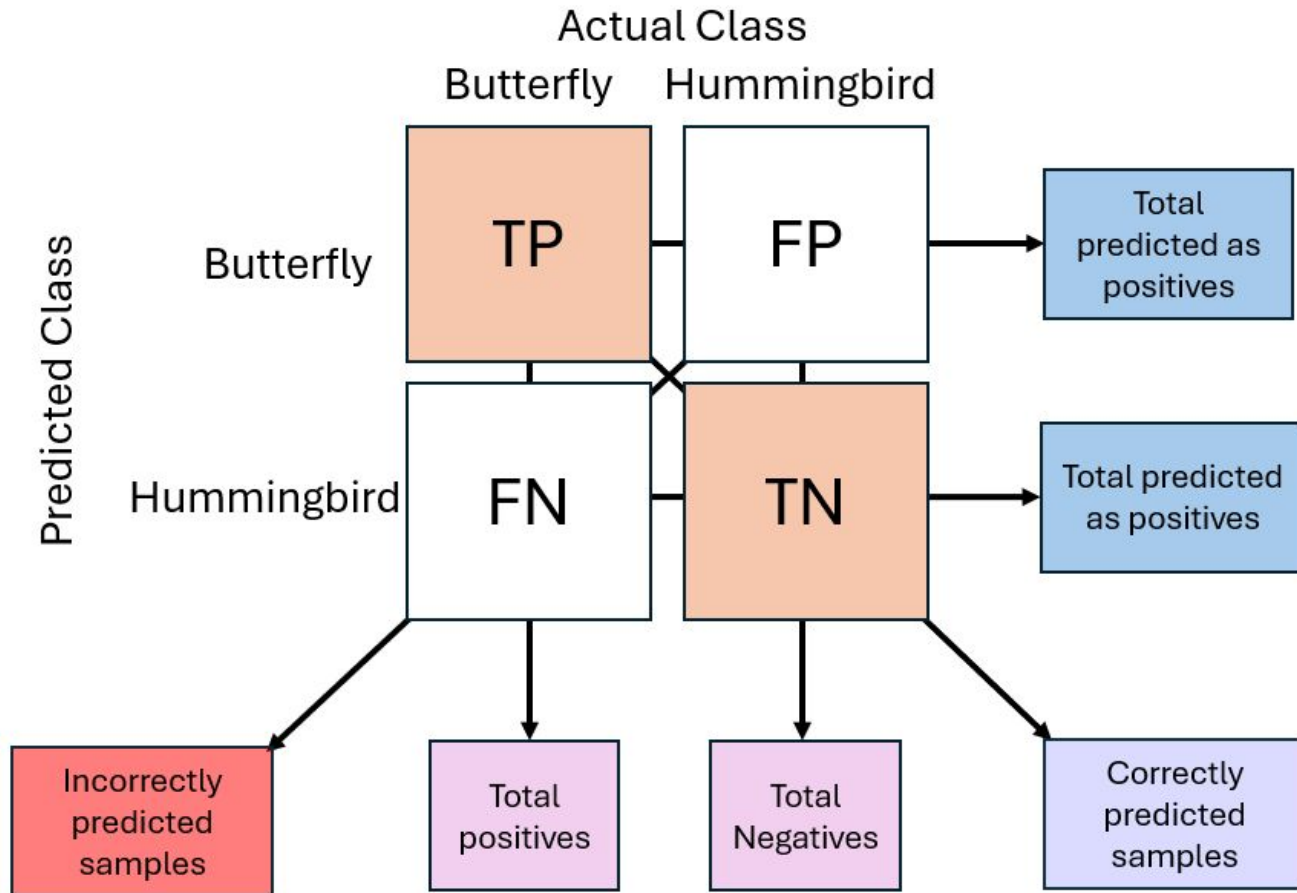
- Accuracy is always a number between 0 and 1, or between 0% and 100%.
- An accuracy of 100% means the model got every prediction correct.
- An accuracy of 0% means the model got every prediction wrong.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Number of total predictions}}$$

Group Work

- Calculate the accuracy of the model based on both the green and orange classifiers.
- Come up with a relationship between TP,FP,FN,TN and accuracy.





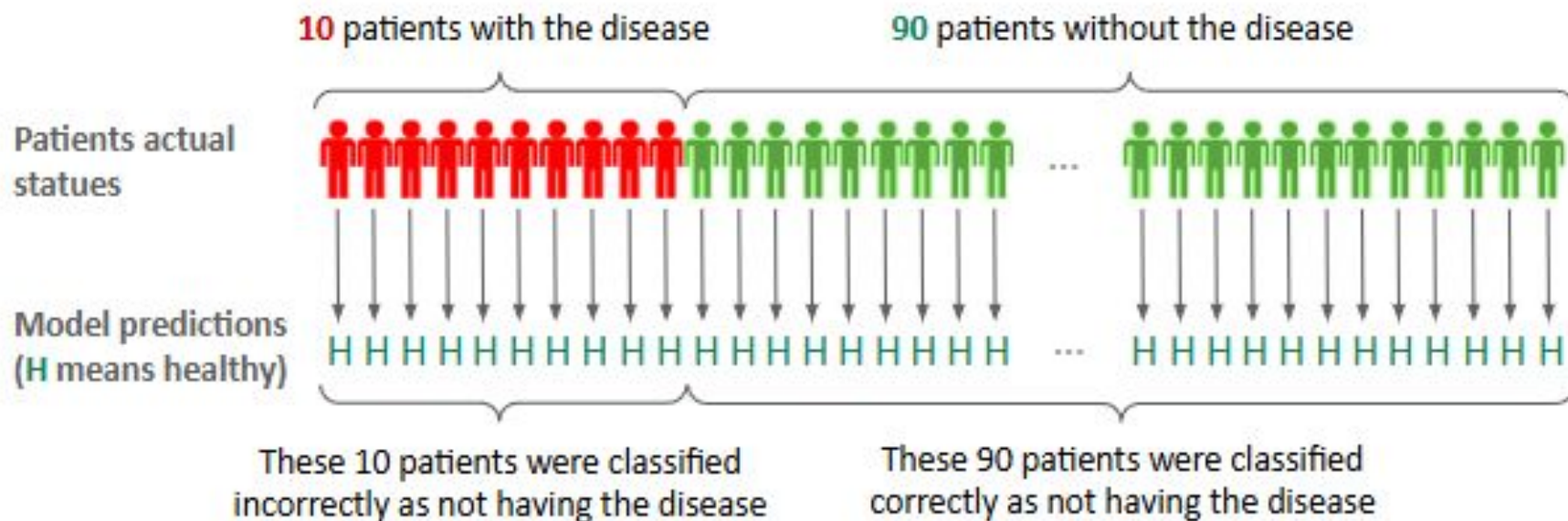
Accuracy

High accuracy does not always mean the model is performing well.

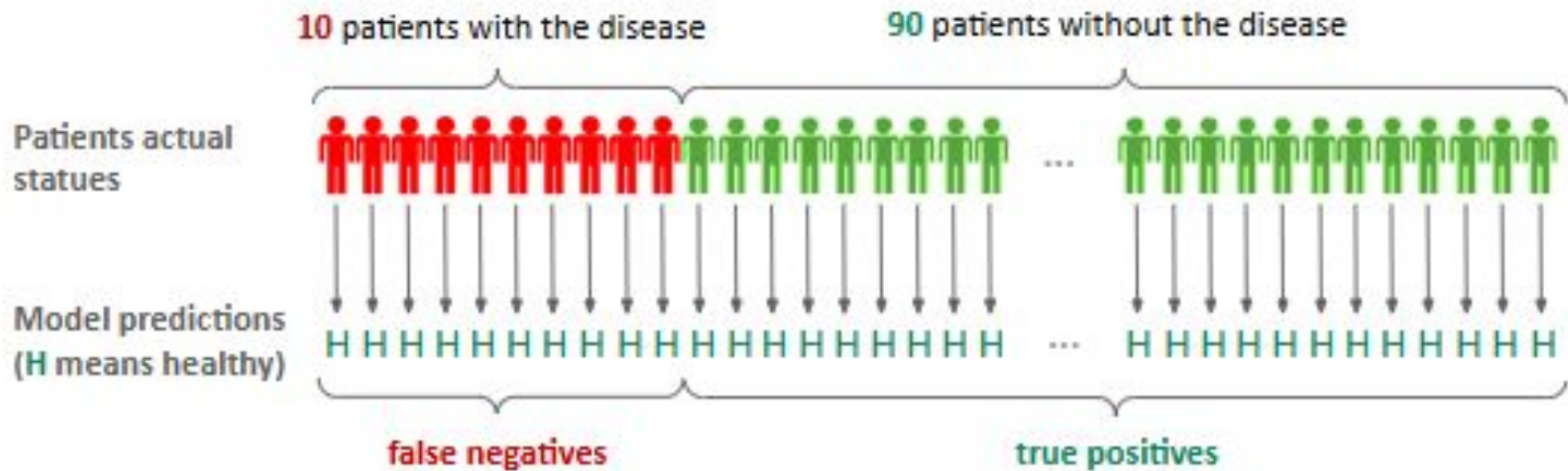
Describe a situation where a model achieves 90% accuracy but still makes poor or meaningless predictions.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Number of total predictions}} = \frac{TP+TN}{TP+FN+FP+TN}$$

Accuracy is not enough



Accuracy is not enough



Accuracy

- While the accuracy is 90%, all predictions on unhealthy patients are false negatives (FN). The model did not make any true positive predictions.
- In class-imbalanced settings we need additional metrics.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Number of total predictions}} = \frac{TP+TN}{TP+FN+FP+TN}$$

Accuracy

- **Pros:** accuracy is simple, intuitive, and gives a quick summary of the model's performance in a single number.
- **Cons:** accuracy can be misleading when the data is imbalanced, especially if one class is much more frequent than the other. it also does not provide detailed insight into how well the model performs in each class.

Recall

- Recall is the fraction of positive examples that were correctly predicted (i.e., classified as positive) over the total number of positive examples (Actual positives).
- Recall is also known as sensitivity and true positive rate (TPR).

$$\text{Recall} = \text{sensitivity} = \text{TPR} = \frac{\text{Number of true positive predictions}}{\text{Number of positive examples}}$$

Individual Work

Deduce the formula for recall using the confusion matrix components (TP, FP, FN, TN).

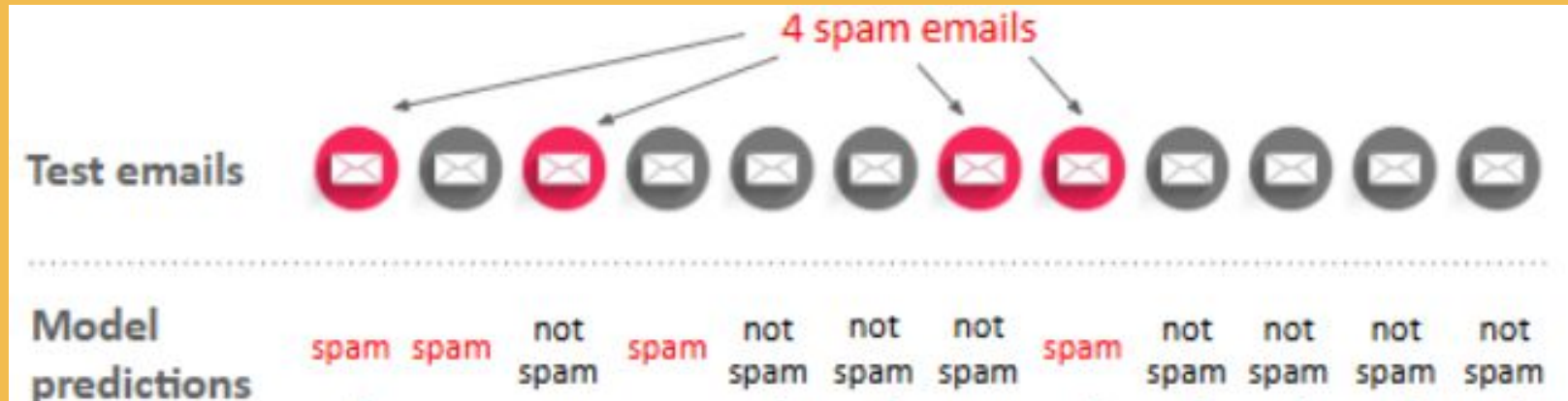
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Recall

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$$\text{Recall} = \text{sensitivity} = \text{TPR} = \frac{\text{Number of true positive predictions}}{\text{Number of positive examples}} = \frac{TP}{TP+FN}$$

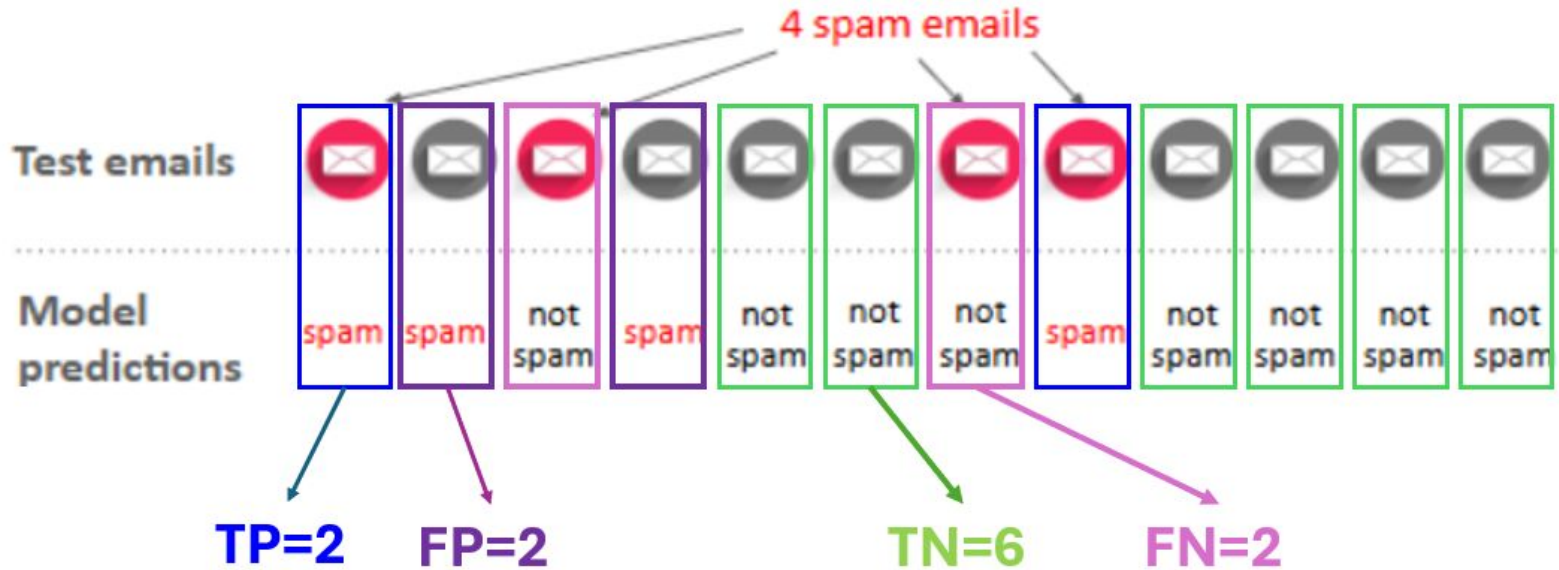
Handout



Guidelines

1. How many are actually spam? How many are actually not spam?
2. How many are predicted as spam? How many are predicted as non-spam?
3. How many are actually spam and were predicted as spam? What do we call this value?
4. How many are actually spam and were predicted as not spam? What do we call this value?
5. How many are actually not spam and were predicted as not spam? What do we call this value?
6. How many are actually not spam and were predicted as not spam? What do we call this value?
7. Is this data imbalanced? explain
8. Construct the confusion matrix.
9. What is the accuracy?
10. What is the recall?

Answer



Answer

		Actual Class	
		Spam	not spam
Predicted Class	Spam	2	2
	not spam	2	6

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} = \frac{\text{diagonal}}{\text{total}} = \frac{8}{12} = 66.66\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{4} = 50\%$$

Recall: Properties

- Recall is always between 0 and 1 (alternatively, from 0% to 100%).
- Recall=100% implies that the model identified all positive examples in the test dataset.
- Recall=0% implies that the model did not identify any of the positive examples.

Group Work

Find

- The confusion matrix of each
- The accuracy of each
- The Recall/Sensitivity/ TPR of each

Test emails												
Model #1 predictions	spam	spam	not spam	spam	not spam	not spam	not spam	spam	not spam	not spam	not spam	not spam
Model #2 predictions	spam	spam	not spam	spam	not spam	not spam	spam	spam	not spam	not spam	not spam	not spam
Model #3 predictions	spam	spam	spam	spam	not spam	not spam	spam	spam	not spam	not spam	spam	spam

Answer

Model 1

		Actual Class	
		Spam	Non spam
Predicted Class	Spam	2	2
	Non-spam	2	6

Accuracy = $8/12=66.67\%$

Recall= $2/4=50\%$

Model 2

		Actual Class	
		Spam	Non spam
Predicted Class	Spam	3	2
	Non-spam	1	6

Accuracy = $9/12=75\%$

Recall= $3/4=75\%$

Model 3

		Actual Class	
		Spam	Non spam
Predicted Class	Spam	4	4
	Non-spam	0	4

Accuracy = $8/12=66.67\%$

Recall = $4/4=100\%$

Session 4: Evaluating Model Performance - Continued

The objectives of the session:

- Discuss the applications of evaluation metrics and provide examples for each use case.
- Explain the purpose and importance of splitting data in machine learning.
- Identify different methods of data splitting, focusing on random splits.
- Distinguish between training, validation, and testing sets and their roles in model development and evaluation.

Examine the terms and definitions provided in this chapter following the [link](#).

Accuracy

Useful when classes are balanced and both false positives and false negatives are equally important.

Example: Quality control in manufacturing where defects and correct products occur in similar proportions.

Recall (Sensitivity, True Positive Rate)

Useful when it's important to capture as many positives as possible, even if some false positives occur.

Example: Disease screening — better to flag all potential cases for further testing than to miss a sick patient.

Specificity (True Negative Rate)

Useful when it's important to correctly identify negatives and avoid false alarms.

Example: Medical tests were mislabeling healthy patients as sick can cause unnecessary stress or treatment.

Precision (Positive Predictive Rate)

Useful when false positives are costly or harmful.

Example: Spam filters — ensuring that emails marked as spam are truly spam, so important emails aren't lost.

Negative Predictive Value (NPV)

Useful when false negatives are costly or harmful.

Example: Safety inspections — ensuring items labeled as “safe” are actually safe to use.

False Alarm Rate

Useful for understanding how often a model incorrectly flags a negative case as positive.

Example: In fraud detection, a high false alarm rate can overwhelm investigators with non-fraud cases.

FPR is the complement of Specificity.

$$FPR = \frac{FP}{FP+TN} = 1 - \textit{Specificity}$$

Individual Work

Find the accuracy, recall, specificity, precision, NPV, and FPR (false alarm rate) for the question in the previous homework.

		Classifier 1	
		Actual Class	
		Spam	Non spam
Predicted Class	Spam	11	4
	Non-spam	4	11

		Classifier 2	
		Actual Class	
		Spam	Non spam
Predicted Class	Spam	10	3
	Non-spam	5	12

Answer

Classifier 1

		Actual Class	
		Spam	Non spam
Predicted Class	Spam	11	4
	Non-spam	4	11

$$\text{Accuracy} = \frac{TP+TN}{TP+FN+FP+TN} = \frac{22}{30} = 73.33\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{11}{15} = 73.33\%$$

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{11}{15} = 73.33\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{11}{15} = 73.33\%$$

$$\text{NPV} = \frac{TN}{TN+FN} = \frac{11}{15} = 73.33\%$$

$$\text{FPR} = \frac{FP}{TN+FP} = \frac{4}{15} = 26.67\%$$

Classifier 2

		Actual Class	
		Spam	Non spam
Predicted Class	Spam	10	3
	Non-spam	5	12

$$\text{Accuracy} = \frac{TP+TN}{TP+FN+FP+TN} = \frac{22}{30} = 73.33\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{10}{15} = 67.67\%$$

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{12}{15} = 80\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{10}{13} = 76.92\%$$

$$\text{NPV} = \frac{TN}{TN+FN} = \frac{12}{17} = 70.59\%$$

$$\text{FPR} = \frac{FP}{TN+FP} = \frac{3}{15} = 20\%$$

Data Splitting

Training set: The largest part of the data that the model learns from.

Validation set: A smaller part of the data used to check how well the model is doing during learning and to help choose the best model.

Testing set: A separate part saved to check how well the model works on completely new data, at the very end.

The Three Pillars of Machine Learning Datasets

THE DEVELOPMENT STAGE

Training Set: Learning Patterns

The model adjusts internal weights to minimize loss and identify general patterns.

Dataset Split	Primary Action	Data Influence
Training	Parameter Adjustment	Direct learning through weight updates
Validation	Hyperparameter Tuning	Indirectly guides model selection decisions

THE FINAL ASSESSMENT

Test Set: Real-World Proxy

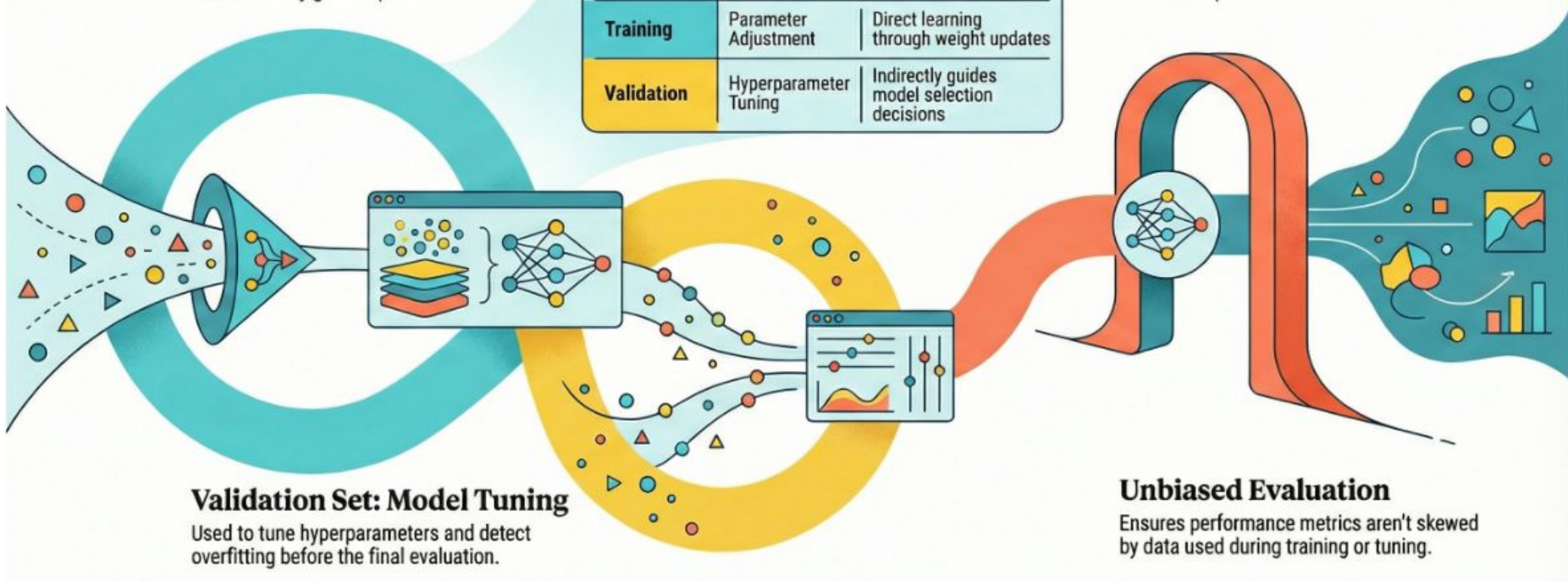
Used only once to provide an unbiased estimate of performance on new data.

Validation Set: Model Tuning

Used to tune hyperparameters and detect overfitting before the final evaluation.

Unbiased Evaluation

Ensures performance metrics aren't skewed by data used during training or tuning.



Pair Work

Discuss and answer the following questions:

1. Why is it important to keep testing data separate from training data?
2. What might happen if we evaluate the model on training data?
3. How would you split a dataset if you only have a small amount of data?

Mastering the Machine Learning Lifecycle: A Step-by-Step Guide

A sequential roadmap from initial data acquisition to final model evaluation.



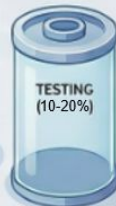
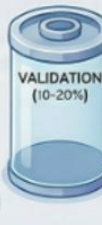
1. Data Collection

Gathering a comprehensive set of labeled images to serve as the foundation for the model.



2. Preprocessing

Data is refined through resizing, normalization, and the removal of blurry images; extracting key features.



3. Data Splitting

The prepared dataset is divided into three distinct sets: training, validation, and testing.



4. Checking Labels

A quality control step where labels are verified for accuracy to present the model from learning incorrect information.



5. Choosing a Model

Developers decide on the specific type of architecture, such as a supervised classifier, based on the project goals.



6. Training

This is the core computational phase where the chosen model is fitted to the training data.



7. Validation

The model is fine-tuned by adjusting hyperparameters to optimize performance and prevent overfitting.



8. Final Testing

The model undergoes a final evaluation using the unseen test images to determine its accuracy and reliability in a real-world scenario.

Pair Work

Write down all the necessary steps to apply the Machine Learning Workflow to Cats and Dogs Data with labeled data (Images tagged as "cat" or "dog")

Answer

Data Collection: Gather a dataset of cat and dog images, each clearly labeled with the correct animal.

Preprocessing: Resize images to the same size, normalize pixel values, remove blurry or corrupt images, extract features

Data Splitting: Randomly split the labeled images into training, validation, and testing sets.

Checking Labels: Verify that the labels ("cat" or "dog") are accurate and consistent.

Answer

Choosing a Model: Select a supervised learning model designed to classify images.

Training: Use the training set and labels to teach the model to recognize patterns distinguishing cats and dogs.

Validation: Check model performance on the validation set to tune settings or stop training at the right time.

Final Testing: Evaluate the trained model on the testing set, which it has never seen before, to estimate how well it will perform on new images.

Pair Work

Write down all the necessary steps to apply the Machine Learning Workflow to Cats and Dogs Data with Unlabeled Data (Images without "cat" or "dog" labels)

Answer

Data Collection: Gather a dataset of cat and dog images with no labels.

Preprocessing: Resize images to a consistent size, normalize pixel values, remove low-quality images, extract features

Data Splitting: Split the images randomly into sets for training and testing (labels are not used here).

Checking Labels: Since there are no labels, this step is skipped or replaced by verifying image quality.

Answer

Choosing a Model: Select an unsupervised learning model (e.g., clustering algorithm) that groups similar images together without labels.

Training: The model tries to find natural groupings or patterns in the images (e.g., grouping similar fur textures or shapes).

Validation: May involve visual inspection of groups or using proxy metrics since no labels are available.

Final Testing: Test how well the groups correspond to meaningful categories (cats vs. dogs) by manual inspection or other indirect methods.

Session 5-6: Overall Process of ML using Python: Practical Session

The objectives of the session:

- Apply learned concepts using coding in Python

Examine the terms and definitions provided in this chapter following the [link](#).

Pair Work

We will start with creating a simple dataset representing weights of two animals—cats and dogs—and then move through important steps in machine learning: splitting the data into training, validation, and testing sets, training a simple model using a threshold, and evaluating its performance using various metrics such as accuracy, recall, precision, and more.

This hands-on exercise will help you see how all these concepts come together in a real workflow.

Follow the [link](#).

Session 7: Summary

The objectives of the session:

- Describe learning approaches for classification, clustering and regression.
- Describe the purpose of distance measures.
- Identify the purpose of cost functions in regression and classification.

Examine the terms and definitions provided in this chapter following the [link](#).

Classification vs Clustering

Goal: Separate data into categories.

Step-by-step idea:

1. Plot data points in feature space.
2. Try to draw a line that separates the classes.
3. If a straight line cannot separate them, try a curved boundary.
4. Adjust the boundary so that it separates classes as well as possible.

The learning process is about finding the best separating boundary between groups.

Classification vs Clustering

Goal: Group similar data without labels.

Step-by-step idea:

1. Measure similarity between points.
2. Group points that are close to each other.
3. Form boundaries around dense regions.
4. Separate clusters based on natural structure in the data.

Clustering finds natural group boundaries without predefined labels.

Learning Approaches for Regression

Goal: Predict a continuous value.

Step-by-step idea:

1. Plot input versus output values.
2. Draw a line that passes close to the data points.
3. If a line is not enough, use a curve.
4. Adjust the line or curve to reduce prediction error.
5. Use the fitted curve to predict future or unseen values.

Regression is about finding the best curve that captures trends in data.

Distance Measures Are the Core of Machine Learning

Most learning methods rely on measuring similarity or difference.

Key idea:

- If two points are close, they are similar.
- If two points are far, they are different.

Distance:

- Guides classification boundaries.
- Forms clusters.
- Influences best-fit calculations.
- Determines error in regression.

Without distance, there is no notion of similarity or error.

The Best Fit Concept and the Objective Function

Best fit means:

- Find the boundary or curve that represents the data most accurately.

To define "best," we create:

- Objective Function = A mathematical measure of error.

Step by step:

1. Make a prediction.
2. Compare prediction to true value.
3. Measure the error.
4. Add all errors together.
5. Adjust parameters to reduce total error.



Objective Function in Classification and Regression

In Regression

We minimize prediction errors.

Common idea:

Minimize the difference between predicted and actual values.

Typically:

- Minimize squared error
- Minimize absolute error

Goal: Make predictions as close as possible to true values.

In Classification

We minimize classification mistakes.

Common idea:

- Reduce misclassification.
- Reduce probability of wrong decisions.
- Increase separation between classes.

Goal: Make the boundary separate classes with minimum error.



Evaluating Models and Why Accuracy Is Not Enough

Evaluating Models and Why Accuracy Is Not Enough

Accuracy = Correct predictions / Total predictions.

But accuracy can be misleading.

Example:

If 95% of data belongs to one class, a model predicting only that class gives 95% accuracy, but it is useless.

Evaluating Models and Why Accuracy Is Not Enough

Evaluation depends on the problem.

Sometimes more important:

- Detecting rare events.
- Avoiding false alarms.
- Identifying critical failures.

Therefore, evaluation must match the application goal.

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